CS-114

Programming Fundamentals

By

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Course

COURSE: FUNDAMENTALS OF PROGRAMMING
COURSE CODE: 114

PRE-REQUISITE: NIL

INSTRUCTOR: ALVINA ANJUM

CREDITS: 3 (2+1)

SEMESTER: SPRING 2026

NUST
Defining futures

Recommended Text Books

- **Recommended Text Books:**
  - **Introduction to Programming with C++, Liang, Y.D, 3rd Edition**
- **Reference Books:**
  - **OOP using C++ by Tasleem Mustafa, Imran Saeed**
  - **Turbo C++ by Robert Lafore.**
  - **C++ How to Program by Deitel**

Course Structure

- **Course Work**
  - Lectures
  - Assignments
  - Quizzes
  - Project

Evaluation Criteria

S. No.	Assessment Items	%age
1	Assignments	10
2	Quizzes / Class Participation	10
3	MID TERM exam (after 7 weeks)	30
4	Final Exams (after 16 weeks)	50

Today's Lecture

- Computers
- Components of computers
  - Hardware
  - Software

Computer



Computer Definition

1. A Computer is an electronic machine which is used for data processing.

2. A computer is a programmable machine, i.e., A machine that can receive, store, and process data.

Computer Definition

3. An electronic device for storing and processing data, typically in binary form, according to instructions given to it in a variable program.

4. A device that computes, especially a programmable electronic machine that performs high-speed mathematical or logical operations according to a set of instructions and that assembles, stores, correlates information.

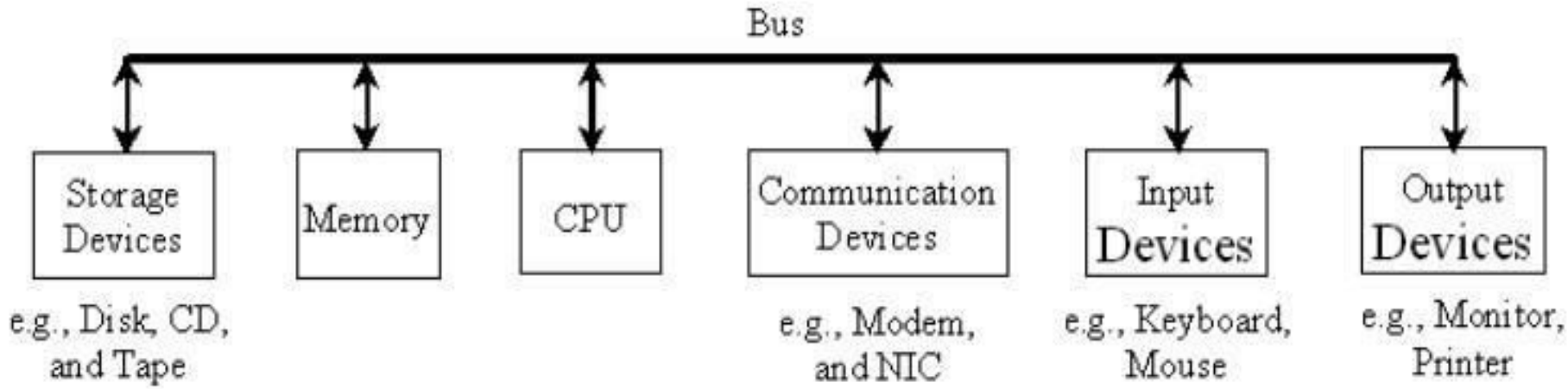
Main Characteristics

- Take input
- Computes
- Programmable electronic machine
- High speed
- Store Information
- Accept Set of instructions
- Produces output

Computer Components

- Two components
  - Hardware
    - The physical equipment required to create, use, manipulate and store electronic data
  - Software
    - Set of instructions written in some programing language
    - The computerised instructions that operate a computer, manipulate the data and execute particular functions or tasks.

What is a Computer System?



Hardware

- Physical parts
  - Memory (Main/ Secondary).
  - CPU(Central Processing Unit)
    - ALU(Arithmetic and Logic Unit)
    - CU(Control Unit).
  - I/O Devices
  - Peripheral device

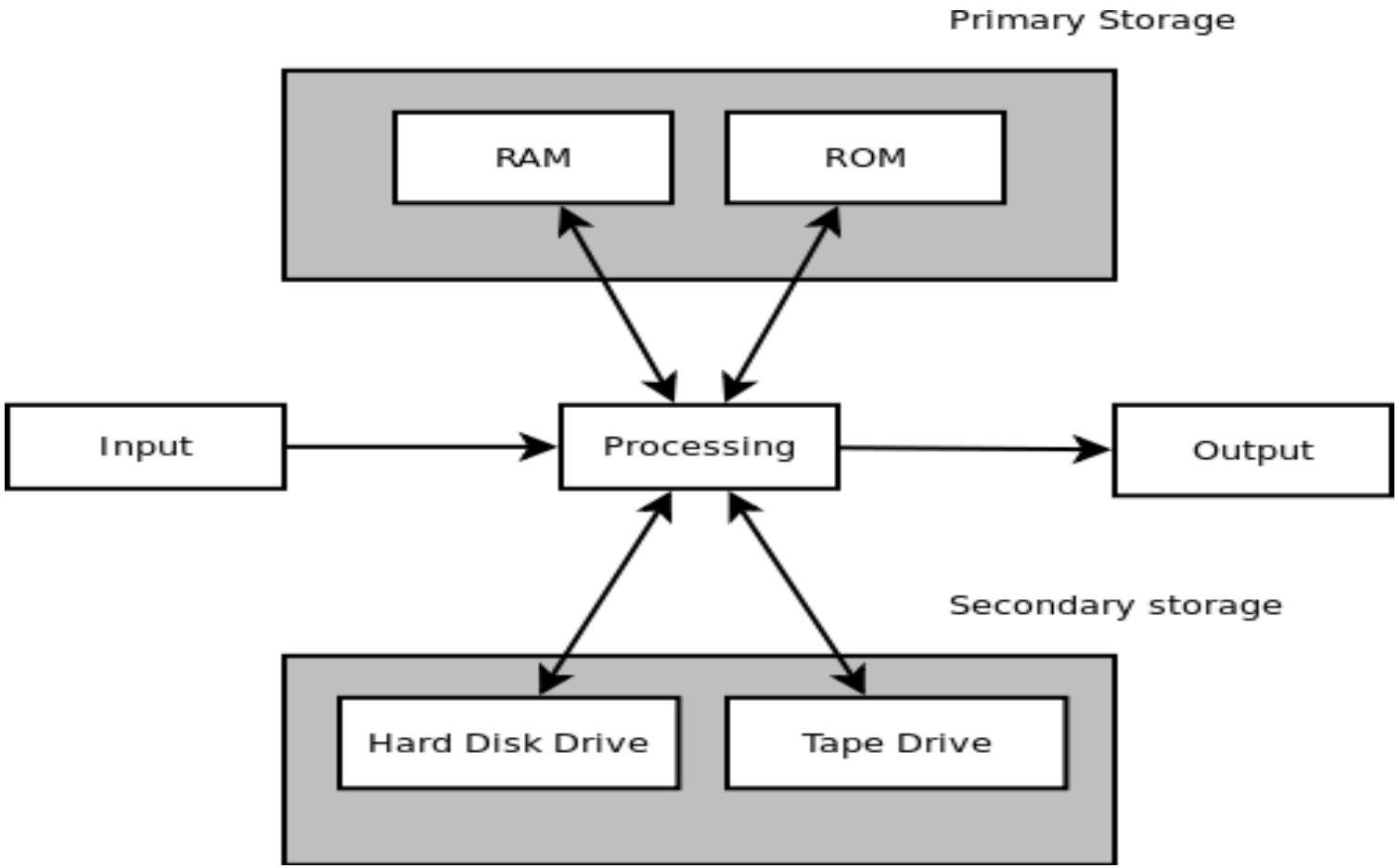
Hardware

- Central processing unit (CPU):
  - The chip or chips at the heart of a computer that enable it to process data.
  - Also known as a processor
- Memory:
  - An area within a computer system that holds data waiting to be processed.
- Storage device:
  - The place where a computer puts data.

Hardware

- Input devices
  - The devices that allow data and instructions to enter a computer (such as a keyboard, mouse, scanner)
- Output devices:
  - the devices that allow information to be represented (that is, given out) to the user, such as a display screen or printer)
- Peripheral device:
  - Any piece of equipment in a computer system that is not actually inside the computer itself.
  - Printers, scanners and external disk drives

Storage Hardware



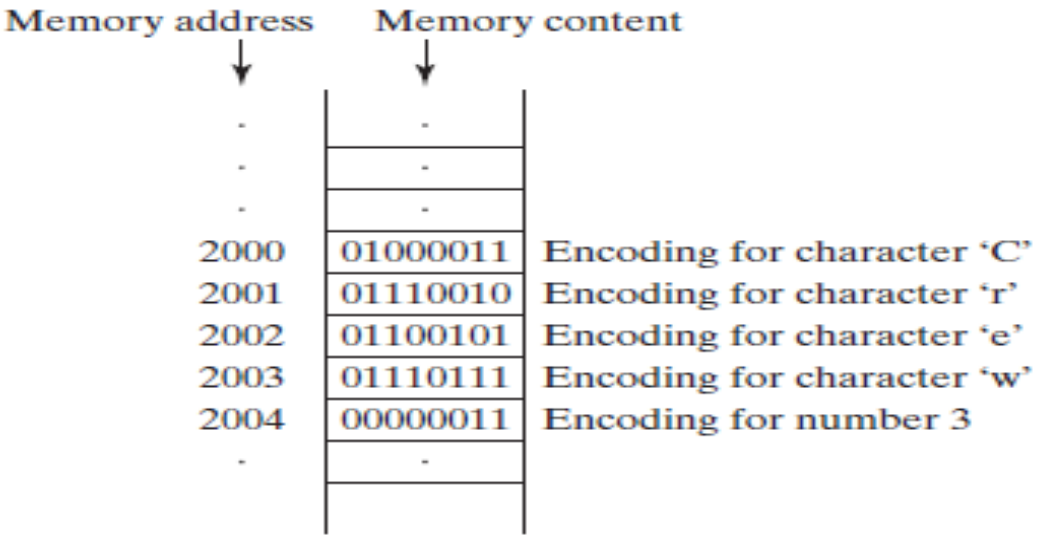
RAM

- The most common type of memory that most users are familiar with is ‘main memory’ or ‘RAM’ (random-access memory).
- RAM holds an area in the computer system unit that temporarily holds a user’s data, operating system instructions and program instructions.
- The word ‘main’ is used to distinguish it from external mass storage devices such as the hard drive or disk drives.



Memory

- **A computer's *memory* consists of an ordered sequence of bytes for storing programs as well as data that the program is working with.**
- **You can think of memory as the computer's work area for executing a program. A program and its data must be moved into the computer's memory before they can be executed by the CPU.**
- **Every byte in the memory has a *unique address*, as shown in Figure**
- **The address is used to locate the byte for storing and retrieving the data. Since the bytes in the memory can be accessed in any order, the memory is also referred to as *random-access memory (RAM)*.**



ROM

Read Only Memory
Memory on which data has been prerecorded. Once data has been written onto a ROM chip, it cannot be removed and can only be read.

- Unlike main memory (RAM), ROM retains its contents even when the computer is turned off.

ROM is referred to as being nonvolatile, whereas RAM is volatile.

- Most PCs contain a small amount of ROM that stores critical program such as the program that boots the computer.

Software

1. Set of instructions that tells a computer component (hardware) how to perform a task.

- Set of instruction = Program

- Program is a set of instructions in any computer programming language which directs computer to do specific tasks.

Software

1. Integrated set of programs
2. Program + data + documentation
3. Simulation of business processes

Types of Software

- There are two types of software
  - System Software
  - Application Software.

System Software

- System software – programs to control the operations of computerequipment
  - Operating system
  - Compiler
  - Loader
  - Linker
  - Interpreter
  - Etc
- System software is a term referring to any computer software which manages and controls the hardware so that application software can perform a task

Application Software

- Application Software includes programs that do real work for user.
  - Manage student database,
  - Payroll systems,
  - Inventory Control,
  - Word Processor, Spreadsheet and
  - Database Management System etc.,
- Application Software – programs that tell a computer how to produce information

Assignment # 1

1. Briefly describe difference between compiler and interpreter.
2. What are application software and system software, distinguish between them with examples.

- Submission date : Next Class

Next Lecture

Why Programming ?